ENR 1.8 REGIONAL SUPPLEMENTARY PROCEDURES

1 All weather operations including low visibility procedures In terms of the Rules of the Air, General Operating Rules, Regulation 91.06.27 paragraphs 1 and 2, no pilot may conduct an instrument approach procedure from a facility unless a State approved instrument procedure has been promulgated for that facility.

1.1 Abbreviations (see appendix A)

1.2 Glossary of Terms

1.2.1 Aerodrome operating minima.

The limits of usability of an aerodrome for either take-off or landing, usually expressed in terms of visibility (VIS) or runway visual range (RVR), decision altitude/height or minimum descent altitude/height and cloud conditions.

1.2.2 All weather operations

Any take-off, en-route or landing operations in IMC and operated in accordance with IFR.

1.2.3 Alternate aerodrome for take-off.

An aerodrome to which a flight may proceed should the weather conditions at the aerodrome of departure preclude a return for landing. Such an aerodrome shall lie within the flying times given below, calculated at normal cruising speed in still air with a critical engine inoperative or critical systems on single engine aircraft becoming inoperative.

Number of Engines	Number of Engines
1	20 Minutes
2	1 Hour
3 or more	2 Hours

1.2.4 Alternate aerodrome for landing.

An aerodrome specified in a flight plan to which a flight may proceed when it becomes inadvisable to land at the aerodrome of intended landing.

1.2.5 Cloud break procedure.

A Cloud Break Procedure may be approved by the Director of Civil Aviation when one or more of the following conditions apply:

a) The runway and/or equipment does not comply with ICAO requirements for instrument approach procedures.
b) The runway is not aligned with the final approach track.
c) Local QNH may not be available. Note: Any cloudbreak procedure conducted outside Controlled Airspace is conducted entirely at the discretion of the Pilot-in-Command. In the event that such runway and/or equipment does not comply with ICAO requirements the OCA/H will not be lower than that specified for circling minima as applicable to aircraft category.

1.2.6 Approach categories of aeroplanes.

The following five approach categories of aeroplanes have been established on the basis of 1.3 times the stall speed in the landing configuration at maximum certificated landing mass:

CATEGORY A - Less than 91 KTS IAS. CATEGORY B - 91 KTS or more, but less than 121 KTS IAS. CATEGORY C - 121 KTS or more, but less than 141 KTS IAS. CATEGORY D - 141 KTS or more, but less than 166 KTS IAS. CATEGORY E - 166 KTS or more, but less than 211 KTS IAS.

1.2.7 Circling approach.

An extension of an instrument approach procedure to provide for visual circling of the aerodrome prior to landing, and ensuring obstacle clearance by restricting the radius of the circling approach as per ICAO document 8168.The visual manoeuvring area for a circling approach is determined by drawing arcs centred on each runway threshold and joining these arcs with tangent lines. The radius of the arc is related to:

a) Aircraft approach category;
b) Speed for each approach category;
c) Wind speed 25 KTS throughout the turn; and
d) Bank angle: 20 average or 3 /second, whichever requires less bank.

Determination method:

The radius is determined using the formulae as per attachment IIIE of ICAO document 8168 by applying 25 kt wind to true airspeed for each category of aircraft using visual manoeuvring IAS from table 1-1 of ICAO document 8168. The TAS is based on an aerodrome elevation of 2000 feet and a temperature of ISA + 15°C.

Example:

For determining radius for visual manoeuvring (circling) area for aerodromes above 2 000 feet MSL.

Approach category of aircraft/ IAS(KTS)	A/100	B/135	C/180	D/205	E/240
TAS at 2000 feet MSL + 25 KTS wind factor (KTS)	131	168	215	242	279
Radius (R) of turn (NM)	0.69	1.13	1.85	2.34	3.12
Category of aircraft/ IAS(KTS)	A/100	B/135	C/180	D/205	E/240
Straight segment (NM) this is a constant value	0.30	0.40	0.50	0.60	0.70
Radius (R) from threshold (NM)	1.68	2.66	4.20	5.28	6.94

Note 1: Radius from threshold (R) = 2 r + straight segment. Note 2: Circling visibility will not be less that approach visibility minima.

1.2.8 Cloud ceiling.

The height above the ground or water of the base of the lowest layer of cloud situated below 20 000 feet and covering more than half the sky.

1.2.9 Decision altitude/height (DA/H).

A specified altitude or height in a precision approach at which a missed approach is initiated if the required visual reference to continue the approach, has not been established.

Note: Decision altitude (DA) is referenced to Mean Sea Level (MSL) and Decision Height (DH) is referenced to the runway threshold elevation.

1.2.10 En-Route safe altitude.

An altitude which will ensure a separation height of at least 1 500 feet above the highest obstacle located within five nautical miles of the aircraft in flight.

1.2.11 Expected approach time (EAT).

a) The time at which ATC expects that an arriving aircraft, following a delay, will leave the holding pattern to complete its approach for a landing.

Note: The actual time of leaving the holding pattern will depend upon the approach clearance.

b) Onward Clearance Time (OCT).

In spite of measures introduced to avoid airborne delays, weather, emergencies and other unpredictable or controllable events may necessitate en-route holding. Pilots are informed that OCT's will only be issued if the hold is expected to exceed 10 minutes.is expected to exceed 10 minutes.

1.2.12 Flight visibility.

The visibility forward from the cockpit of an aircraft in flight.

1.2.13 Final approach fix or point.

That fix or point of an instrument approach procedure where the final approach segment commences.

1.2.14 Initial approach fix.

The fix determined in terms of Instrument Approach Procedures which identifies the beginning of the initial approach segment.

1.2.15 Ground controlled approach (GCA).

A ground radar control service which encompasses all facets of the radar control service in the vicinity of a SAAF airbase i.e., vectoring, terrain clearance and radar final approach service.

No civilian pilot may conduct the final approach phase of a Ground Controlled Approach Procedure (GCA) consisting of the continuous talk down phase.

Note: The GCA is not approved by the Director of for Civil Aviation.

1.2.16 Initial approach altitude.

The lowest altitude in the holding pattern or on an arrival route of an instrument approach procedure, providing the required protection against obstacles. This altitude may be below Minimum Sector Altitude (MSA) and accepted by a pilot when being vectored under radar control.

1.2.16.1 Procedure turn approach.

A procedure turn approach may, with ATC approval where an ATS is operational be executed if approaching the navigational aid within 30DEG from the outbound track.

1.2.17 Instrument approach procedure.

A series of predetermined manoeuvres by reference to flight instruments with specified protection from obstacles from the initial approach fix, or where applicable, from the beginning of a defined arrival route, to a point from which a landing can be completed and thereafter, if a landing is not completed, to a position at which holding or en-route obstacle criteria apply.

1.2.18 Minimum descent altitude/height (MDA/H).

A specified altitude or height in a non-precision approach or circling approach below which descent may not be made without visual references for the intended runway or touch-down area.

Note: Minimum Descent Altitude (MDA) is referenced to Mean Sea Level (MSL) and Minimum Descent Height (MDH) is referenced to runway threshold or aerodrome elevation.

1.2.19 Minimum sector altitude (MSA).

The lowest altitude which may be used which will provide a minimum clearance of 1 500 ft (450 m) above all objects located in an area contained within a sector of a circle of 25 NM (46 km) radius centred on a radio aid to navigation significant point, the Aerodrome Reference Point (ARP) or the Heliport Reference Point (HRP).

Note: It is the responsibility of the pilot-in-command not to descend prematurely below MSA, unless being vectored under radar control.

1.2.20 Missed approach point (MAPt).

That point in an instrument approach procedure at or before which the prescribed missed approach procedure must be initiated in order to ensure that the minimum obstacle clearance is not infringed.

1.2.21 Missed approach procedure.

The procedure to be followed if the approach cannot be continued.

1.2.22 Non-precision approach.

a) An instrument approach for landing in which no electronic glide path guidance is provided.

b) In the case where there are two facilities utilised for the instrument approach procedure, the facility on which the instrument approach procedure is based shall be deemed to be the primary facility and the second facility shall be deemed to be the secondary facility.

c) In circumstances where the secondary facility is not available whilst executing an instrument approach procedure and the secondary facility, when operative, would have provided a lower MDA, then the aircraft may descend on a time basis to the specified altitude for the secondary facility, unless a different MDA/H has been specified in the event of such facility not being available.

1.2.23 Obstacle clearance altitude (OCA) or obstacle clearance height (OCH).

The lowest altitude or the lowest height above the elevation of the relevant runway threshold or the aerodrome elevation as applicable, used in establishing compliance with appropriate obstacle clearance criteria.

Note 1. - Obstacle clearance altitude is referenced to mean sea level and obstacle clearance height is referenced to the threshold elevation or in the case of non-precision approaches to the aerodrome elevation or the threshold elevation if that is more than 2 m (7 ft) below the aerodrome elevation. An obstacle clearance height for a circling approach is referenced to the aerodrome elevation.

Note 2. - For convenience when both expressions are used they may be written in the form obstacle clearance altitude/height and abbreviated OCA/H.

Note 3. - See Procedures for Air Navigation Services, Part I, Section 4, Chapter 5, 5.4 for specific applications of this definition.
Note 4. - See Part IV, Chapter 1 for Area navigation (RNAV) point-in-space (PinS) approach procedures for helicopters using basic GNSS receivers, Part IV, Chapter 1. The general criteria for OCA/H apply (Part I, Section 4, Chapter 5, 5.4) with the addition that the OCH is above the highest terrain/surface within 1.6 km (0.86 NM) of the MAPt.

1.2.24 Obstacle clearance criteria.

The criteria used by the State that provides vertical separation from obstacles to aircraft in flight in various phases of an instrument approach procedure and which under no circumstances will be less than the minima recommended by ICAO in document 8168/ OPS as amended.

1.2.25 Precision approach.

An Instrument Approach for landing in which precision azimuth guidance and precision glide path guidance are provided in accordance with the minima prescribed for the category of operation.

1.2.26 Required visual reference.

The runway environment, i.e. runway threshold, touch-down area, touch down zone lighting, or approach lighting which must be visible for sufficient time for the pilot to determine that the aircraft is in a position for a normal visual descent to land.

1.2.27 State approved procedures.

No pilot may conduct an instrument approach procedure from a facility under any circumstances unless a State approved instrument approach procedure has been promulgated by NOTAM/AIP or AIP SUP action for that facility.

1.2.28 RNAV approach

An instrument approach procedure which relies on aircraft area navigation equipment for navigation guidance.

1.2.29 Runway visual range (RVR).

The range over which the pilot of an aeroplane on the centre line of a runway can see the runway surface markings or the lights delineating the runway or identifying its centre line.

1.2.30 Visibility.

The ability, as determined by atmospheric conditions and expressed in units of distance, to see and identify prominent unlighted objects by day and prominent lighted objects by night.

1.2.31 Visual approach

1) Subject to the conditions in paragraph 3 below, clearance for an IFR flight to execute a visual approach may be requested by a flight crew or initiated by the controller. In the latter case, the concurrence of the flight crew shall be required.
2) Controllers shall exercise caution in initiating a visual approach when there is reason to believe that the flight crew concerned is not familiar with the aerodrome and its surrounding terrain. Controllers should also take into consideration the prevailing traffic and weather conditions when initiating visual approaches.
3) An IFR flight may be cleared to execute a visual approach provided that the pilot can maintain visual reference to the terrain and:

a) the reported ceiling is at or above the approved initial approach ALT for the aircraft so cleared; or
b) the pilot reports at the initial approach level or at any time during the instrument approach procedure that the meteorological conditions are such that with reasonable assurance exists that a a visual approach can be completed; and
c) the aircraft is within 25 NM of the destination aerodrome.Separation shall be provided between an aircraft cleared to execute a visual approach and other traffic in the vicinity of the destination aerodrome.

1.2.32 Visual descent point (VDP).

A defined point on the final approach track of a non-precision straight-in approach procedure from which a normal descent from the MDA to the runway touchdown zone may be commenced, provided the approach threshold of that runway or approach lights or other markings are clearly visible to the pilot.

1.2.33 Clearance to descend subject to maintaining own separation while in visual meteorological conditions.

When requested by the pilot of an aircraft and if so prescribed by the appropriate ATS authority an arriving aircraft may be cleared to descend subject to maintaining own separation from other aircraft and remaining in visual meteorological conditions if reports indicate that this is possible.

2 AIRSPACE DEFINITIONS.

2.1 Advisory airspace:

An airspace of defined dimensions, within which air traffic advisory service is available.

Note:

This airspace may be designated advisory area (ADA) or advisory route (ADR).

2.2 Controlled Airspace.

An airspace of defined dimensions within which an air traffic control service is provided to IFR flights and to VFR flights in accordance with the airspace classification as prescribed in Regulation 172.02.2.

Note:

Controlled airspace is a generic term which covers ATS airspace classes A, B, C, D and E as defined in ICAO Annex 11, appendix 4.

3 AERODROME OPERATING MINIMA

3.1 Take-off minima

3.1.1 General

a) Take-off minima established by the operator must be expressed as visibility or RVR limits, taking into account all relevant factors for each aerodrome planned to be used and the aeroplane characteristics. Where there is a specific need to see and avoid obstacles on departures and/or for a forced landing, additional conditions (e.g. ceiling) must be specified.
b) The pilot-in-command may not commence take-off unless the weather conditions at the aerodrome of departure are equal to or better than the applicable minima for landing at that aerodrome unless a suitable take-off alternate aerodrome is available.
c) When the reported meteorological visibility is below that required for take-off and RVR is not reported, a take-off may only be commenced if the pilot-in-command can determine that the RVR/visibility along the take-off runway is equal to or better than the required minimum.
d) When no reported meteorological visibility or RVR is available, a take-off may only be commenced if the pilot-in-command can determine that the RVR/visibility along the take-off runway is equal to or better than the required minimum.

3.1.2 Visual reference

The take-off minima must be selected to ensure sufficient guidance to control the aeroplane in the event of both a discontinued take-off in adverse circumstances and a continued take-off after failure of the critical power unit.

3.1.3 Required RVR/Visibility

a) For multi-engine aeroplanes whose performance is such that, in the event of a critical power unit failure at any point during take-off, the aeroplane can either stop or continue the take-off to a height of 1500 feet above the aerodrome while clearing obstacles by the required margins, the take-off minima established by an operator must be expressed as RVR/Visibility values not lower than those given in Table 1 below except as provided in paragraph (3.1.4) below:

TABLE 1: RVR / VISIBILITY FOR TAKE-OFF

Take-off RVR / Visibility	
FACILITIES	**RVR/VISIBILITY (Note 3)**
Nil (day only)	500m
Runway edge lighting and/or centre line marking	250/300m (Notes 1 & 2)
Runway edge and centre line lighting	200/250m (Note 1)
Runway edge and centre line lighting and multiple RVR information	150/200m (Notes 1 & 4)
Note 1	The higher values apply to Category D aeroplanes
Note 2	For night operations at least runway edge and runway end lights are required.
Note 3	The reported RVR/visibility value representative of the initial part of the take-off run can be replaced by pilot assessment
Note 4	The required RVR value must be achieved for all of the relevant RVR reporting points with the exception given in Note 3 above.

b) For multi-engined aeroplane whose performance is such that they cannot comply with the performance conditions in subparagraph (3.1.3)(a) above in the event of a critical power unit failure, there may be a need to re-land immediately and to see and avoid obstacles in the take-off area. Such aeroplanes may be operated to the following take-off minima provided they are able to comply with the applicable obstacle clearance criteria, assuming engine failure at the height specified. The take-off minima established by an operator must be based upon the height from which the one engine inoperative net take-off flight path can be constructed. The RVR minima used may not be lower than either of the values given in Table 1 above or Table 2 below.

TABLE 2 : ASSUMED ENGINE FAILURE HEIGHT ABOVE THE RUNWAY VERSUS RVR / VISIBILITY	
TAKE OFF RVR / VISIBILITY - FLIGHT PATH	
Assumed engine failure height above the take-off runway	RVR/Visibility (Note 2)
< 50ft	200m
51 - 100 ft	300m
101- 150 ft	400m
151 - 200 ft	500m
201 - 300 ft	1 000m
> 300 ft	1 500 m (Note 1)
Note 1	1 500 m is also applicable if no positive take-off flight path can be constructed.
Note 2	The reported RVR/Visibility value representative of the initial part of the take-off run can be replaced by pilot assessment.

c) When reported RVR, or meteorological visibility is not available, the pilot-in-command may not commence take-off unless he or she can determine that the actual conditions satisfy the applicable take-off minima.

3.1.4 Exceptions to Paragraph (3.1.3)(a)

a) Subject to approval of the Director, and provided the requirements in paragraphs (i) to (v) below have been satisfied, an operator may reduce the take-off minima to 125 m RVR (Category A, B and C aeroplanes) or 150 m RVR (Category D aeroplanes) when:-
   i) Low visibility procedures are in force;
   ii) high intensity runway center line lights spaced 15 m or less and high intensity edge lights spaced 60 m or less are in operation;
   iii) flight deck crew members have satisfactorily completed training in a simulator approved for this procedure;
   iv) a 90 m visual segment is available from the cockpit at the start of the take-off run; and
   v) the required RVR value has been achieved for all of the relevant RVR reporting points.

b) Subject to the approval of the Director, an operator of an aeroplane using an approved lateral guidance system for take-off may reduce the take-off minima to an RVR less than 125 m (Category A, B and C aeroplanes) or 150 m (Category D aeroplanes) but not lower than 75 m provided runway protection and facilities equivalent to Category III landing operations are available.

3.2 NON-PRECISION APPROACH

3.2.1 System Minima

a) An operator must ensure that system minima for non-precision approach procedures, which are based upon the use of ILS without glide path (LOC only) VOR, NDB, SRA or VDF are not lower than the MDH values given in Table 3 below.

TABLE 3: SYSTEM MINIMA FOR NON-PRECISION APPROACH AIDS

SYSTEM MINIMA	
FACILITY	**LOWEST MDH**
ILS (no glide path - LOC)	250 ft
SRA (terminating at 1\2 NM)	250 ft
SRA (terminating at 1NM)	300 ft
SRA (terminating at 2 NM)	350 ft
VOR	300 ft
VOR/DME	250 ft
NDB	300 ft
VDF	300 ft

3.2.2 Minimum Descent Height

An operator must ensure that the minimum descent height for a non-precision approach is not lower than either -

a) the OCH/OCL for the category of aeroplane; or
b) the system minimum.

3.2.3 Visual Reference

A pilot may not continue an approach below MDA/MDH unless at least one of the following visual references for the intended runway is distinctly visible and identifiable to the pilot:

a) Elements of the approach light system;
b) The threshold;
c) The threshold markings;
d) The threshold lights;
e) The threshold identification lights;
f) The visual glide slope indicator;
g) The touchdown zone or touchdown zone markings;
h) The touchdown zone lights;
i) Runway edge lights; or
j) Other visual references acceptable to the Director.

3.2.4 Required RVR

The lowest minima to be used by an operator for non-precision approaches are:

TABLE 4(a): RVR FOR NON-PRECISION APPROACH-FULL FACILITIES (Notes (1), (5), (6) and (7))				
RVR/Aeroplane category				
MDH	A	B	C	D
250 ft - 299 ft	800 m	800 m	800 m	1 200m
300 ft - 449 ft	900 m	1 000 m	1 000 m	1 400 m
450 ft - 649 ft	1 000 m	1 200 m	1 200 m	1 600 m
650 ft and above	1 200 m	1 400 m	1 400 m	1 800 m

TABLE 4(b): RVR FOR NON-PRECISION APPROACH-INTERMEDIATE FACILITIES				
Non-precision approach minima Intermediate facilities (Notes (2),(5),(6) and (7))				
MDH	**RVR/Aeroplane category**			
	A	**B**	**C**	**D**
250 ft - 299 ft	1 000 m	1 100 m	1 200 m	1 400 m
300 ft - 449 ft	1 200 m	1 300 m	1 400 m	1 600 m
450 ft - 649 ft	1 400 m	1 500 m	1 600 m	1 800 m
650 ft and above	1 500 m	1 500 m	1 800 m	2 000 m

TABLE 4(c): RVR FOR NON-PRECISION APPROACH-BASIC FACILITIEST				
Non- precision approach minima Basic facilities (Notes (3),(5),(6) and (7))				
MDH	**RVR/Aeroplane category**			
	A	**B**	**C**	**D**
250 ft - 299 ft	1 200 m	1 300 m	1 400 m	1 600 m
250 ft - 299 ft	1 300 m	1 400 m	1 600 m	1 800 m
450 ft - 649 ft	1 500 m	1 500 m	1 800 m	2 000 m
650 ft and above	1 500 m	1 500 m	2 000 m	2 000 m

TABLE 4(d): RVR FOR NON PRECISION APPROACH-NIL APPROACH LIGHT FACILITIES				
Non-precision approach minima Nil facilities (Notes (4),(5),(6) and (7))				
MDH	**RVR/Aeroplane category**			
	A	**B**	**C**	**D**
250 ft - 299 ft	1 500 m	1 500 m	1 600 m	1 800 m
300 ft - 449 ft	1 500 m	1 500 m	1 800 m	2 000 m
450 ft - 649 ft	1 500 m	1 500 m	2 000 m	2 000 m
650 ft and above	1 500 m	1 500 m	2 000 m	2 000 m
Note 1	Full facilities comprise runway markings, 720 m or more of HI/MI approach lights, runway edge lights, threshold lights and runway end lights. Lights must be on.			
Note 2	Intermediate facilities comprise runway markings, 420-719 m of HI/MI approach lights, runway edge lights, threshold lights and runway end lights. Lights must be on			
Note 3	Basic facilities comprise runway markings, <420 m of HI/MI approach lights, any length of LI approach lights, runway edge lights, threshold lights and runway end lights. Lights must be on.			
Note 4	Nil approach light facilities comprise runway markings, runway edge lights, threshold lights, runway end lights or no lights at all.			
Note 5	The tables are only applicable to conventional approaches with a nominal descent slope of not greater than 4. Greater descent slopes will usually require that visual glide slope guidance (e.g. PAPI) is also visible at the Minimum Descent Height.			
Note 6	The above figures are either reported RVR or meteorological visibility converted to RVR as in Technical Standard 121.07.7.8.			
Note 7	The MDH mentioned in Table 4(a), 4(b), 4(c) and 4(d) refers to the initial calculation of MDH. When selecting the associated RVR, there is no need to take account of a rounding up to the nearest ten feet, which may be done for operational purposes, e.g. conversion to MDA.			

3.2.5 Night Operations

At least runway edge, threshold and runway end lights must be on.

3.3 PRECISION APPROACH - CATEGORY 1 OPERATIONS

3.3.1 General

A Category I operation is a precision instrument approach procedure which provides for an approach to a decision height not lower than 200 ft and a visibility not less than 800m or RVR not less than 550m.

3.3.2 Decision Height

An operator must ensure that the decision height to be used for a Category I precision approach is not lower than -

a) the minimum decision height specified in the aeroplane flight manual (AFM) if stated;

b) the minimum height to which the precision approach aid can be used without the required visual reference;
c) the OCH/OCL for the category of aeroplane; or
d) 200 ft

3.3.3 Visual Reference

A pilot may not continue an approach below the Category I decision height determined in accordance with paragraph 2 above, unless at least one of the following visual references for the intended runway is distinctly visible and identifiable to the pilot:

a) elements of the approach light system;
b) the threshold;
c) the threshold markings;
d) the threshold lights;
e) the threshold identification lights;
f) the visual glide slope indicator;
g) the touchdown zone or touchdown zone markings;
h) the touchdown zone lights; or
i) runway edge lights.

3.3.4 Required RVR

The lowest minima to be used by an operator for Category I operations are:

TABLE 5: RVR FOR CAT I APPROACH VS. FACILITIES AND DH

Category I minima

DH	Facilities / RVR (Note 5)			
	Full (Notes1&6)	**Interm (Notes2&6)**	**Basic (notes 3&6)**	**Nil (Notes 4&6)**
200 ft	550 m	700 m	800 m	1 000 m
201-250 ft	600 m	700 m	800 m	1 000 m
251-300 ft	650 m	800 m	900 m	1 200 m
301 ft and above	800 m	900 m	1 000 m	1 200 m
Note 1	Full facilities comprise runway markings, 720 m or more of HI/MI approach lights, runway edge lights, threshold lights and runway end lights. Lights must be on.			
Note 2	Intermediate facilities comprise runway markings, 420 - 719 m of HI/MI approach lights, runway edge lights, threshold lights and runway end lights. Lights must be on.			
Note 3	Basic facilities comprise runway markings, <420 m of HI/MI approach lights, any length of LI approach lights, runway edge lights, threshold lights and runway end lights. Lights must be on.			
Note 4	Nil approach light facilities comprise runway markings, runway edge lights, threshold lights, runway end lights or no lights at all.			
Note 5	The above figures are either the reported RVR or meteorological visibility converted to RVR in accordance with Technical Standard 131.6.			

DH	Facilities / RVR (Note 5)			
	Full (Notes1&6)	**Interm (Notes2&6)**	**Basic (notes 3&6)**	**Nil (Notes 4&6)**
Note 6	The table is applicable to conventional approaches while a glide slope angle up to and including 4.			
Note 7	The DH mentioned in Table 5 refers to the initial calculation of DH. When selecting the associated RVR, there is no need to take account of a rounding up to the nearest ten feet, which may be done for operational purposes, e.g. conversion to DA.			

3.3.5 Single Pilot Operations

For single pilot operations, an operator must calculate the minimum RVR for all approaches in accordance with CAR 121.07.7 and this technical standard. An RVR of less than 800 m is not permitted except when using a suitable auto-pilot coupled to an ILS or MLS, in which case normal minima apply. The decision height applied may not be less than 1.25 X the minimum disengagement height for the autopilot.

3.3.6 Night Operations

For night operations at least runway edge, threshold and runway end lights must be on.

3.4 PRECISION APPROACH - CATEGORY II OPERATIONS

3.4.1 General

A Category II operation is an ILS approach procedure which provides for an approach to a decision height lower than 200 feet but not lower than 100 feet and a RVR of not less than 300m.

3.4.2 Decision Height

An operator must ensure that the decision height for a Category II operation is not lower than -

a) the minimum decision height specified in the AFM, if stated;
b) the minimum height to which the precision approach aid can be used without the required visual reference;
c) the OCH/OCL for the category of aeroplane; or
d) the decision height to which the flight crew is authorised to operate; or
e) 100 ft.

3.4.3 Visual Reference

A pilot may not continue an approach below the Category II decision height determined in accordance with paragraph (3.4.2) above, unless visual references containing a segment of at least 3 consecutive lights being the centre line of the approach lights, or touchdown zone lights, or runway centre line lights, or runway edge lights, or a combination of these is attained and can be maintained. This visual reference must

include a lateral element of the ground pattern, i.e. an approach lighting crossbar or the landing threshold or a barrette of the touchdown zone lighting

3.4.4 Required RVR

The lowest minima to be used by an operator for Category II operations are:

TABLE 6: RVR FOR CATEGORY II APPROACH VS DH		
Category II minima		
Decision height	**Auto-coupled to below DH (Note 1)**	
	RVR/Aeroplane Category A,B & C	**RVR/Aeroplane Category D**
100FT - 120FT	300M	300M (note 2) / 350M
121FT - 140FT	400M	400M
141FT and above	450M	450M
Note 1	The reference to auto-coupled to below DH in this table means continued use of the automatic flight control system down to a height which is not greater than 80% of the applicable DH. Thus airworthiness requirements may, through minimum engagement height for the automatic flight control system, affect the DH to be applied.	
Note 2	300M may be used for a Category D aeroplane conducting an autoland.	

3.5 PRECISION APPROACH - CATEGORY III OPERATIONS

3.5.1 General

Category III operations are subdivided as follows:

a) Category IIIA Operations

An ILS approach procedure which provides for an approach to a decision height lower than 100FT or with no decision height and with a RVR of not less than 200M.

b) Category IIIB operations.

An ILS approach procedure which provides for approach with either decision height lower than 50FT or no decision height and a RVR lower than 200M but not less than 75M.

c) Category IIIC operations.

An ILS approach procedure which provides for approach with no decision height and no runway visual range limitations.

3.5.2 Decision Height

For operations in which a decision height is used, an operator must ensure that the decision height is not lower than:-

a) the minimum decision height specified in the AFM, if stated;
b) the minimum height to which the precision approach aid can be used without the required visual reference; or
c) the decision height to which the flight crew is authorised to operate.

3.5.3 No Decision Height Operations

Operations with no decision height may only be conducted if:-

a) the operation with no decision height is authorised in the AFM;
b) the approach aid and the aerodrome facilities can support operations with no decision height; and
c) the operator has an approval for CAT III operations with no decision height.

Note: In the case of a Category III runway it may be assumed that operations with no decision height can be supported unless specifically restricted as published in the AIP or NOTAM.

3.5.4 Visual Reference

a) For Category IIIA operations, a pilot may not continue an approach below the decision height determined in accordance with paragraph 3.5.2 above unless a visual reference containing a segment of at least 3 consecutive lights being the centre line of the approach lights, or touchdown zone lights, or runway centre line lights, or runway edge lights, or a combination of these is attained and can be maintained.
b) For Category IIIB operations with a decision height a pilot may not continue an approach below the decision height, determined in accordance with paragraph 3.5.2 above, unless a visual reference containing at least one centre line light is attained and can be maintained.
c) For Category III operations with no decision height there is no requirement for visual contact with the runway prior to touch-down.

3.5.5 Required RVR

The lowest minima to be used by an operator for Category III operations are:

TABLE 7: RVR FOR CATEGORY III APPROACH VERSUS FLIGHT CONTROL SYSTEMS AND DH

Category III minima					
		Flight control system/ RVR (metres)			
		Fail - Passive	Fail - Operational		
			Without roll-out system	With roll-out guidance or control system	
Approach category	Decision height (ft)			Fail passive	Fail operational
III A	Less than100FT	175M (Note 1)	175M	175M	175M

TABLE 7: RVR FOR CATEGORY III APPROACH VERSUS FLIGHT CONTROL SYSTEMS AND DH					
III B	Less than50 ft	Not Authorised	Not Authorised	125 m	75 m
III C	No DH	Not Authorised	Not Authorised	Not Authorised	75 m
Note 1	For operations to actual RVR values less than 300 m a go-around is assumed in the event of an autopilot failure at or below DH.				

3.6 CIRCLING

3.6.1 The lowest minima to be used by an operator for circling are:

TABLE 8 VISIBILITY AND MDH FOR CIRCLING VS AEROPLANE CATEGORY				
	Aeroplane Category			
	A	**B**	**C**	**D**
Minimum OCH	295 ft	295 ft	394 ft	394 ft
MDH	400 ft	500 ft	600 ft	700 ft
Minimum meteorological visibility	1 500 m	1 600 m	2 400 m	3 600 m

3.6.2 Circling with prescribed tracks is an accepted procedure within the meaning of this paragraph.

3.7 VISUAL APPROACH

An operator may not use a RVR of less than 1 500m for a visual approach.

3.8 CONVERSION OF REPORTED METEOROLOGICAL VISIBILITY TO RVR

3.8.1 An operator must ensure that a meteorological visibility to RVR conversion is not used for calculating take-off minima, Category II or III minima or when a reported RVR is available.

3.8.2 When converting meteorological visibility to RVR in all other circumstances than those in paragraph (3.8.1) above, an operator must ensure that the following table is used:

TABLE 9: CONVERSION OF VISIBILITY TO RVR		
Lighting elements in operation	**RVR = Reported MET Visibility x**	
	Day	**Night**
HI approach and runway lighting	1.5	2
Any type of lighting installation other than above	1	1.5
No lighting	1	Not applicable

4 LOW VISIBILITY PROCEDURES (LVP's)

4.1 Introduction

These procedures have been devised to simplify the differing requirements of Category II and III operations.To achieve this, during the relevant weather conditions (see 2 below), the localizer sensitive area (LSA) is to be safe guarded. This ensures the protection of the localizer signal against interference and at the same time effectively meets the obstacle free zone (OFZ) requirements.

4.2 General

ATC Low Visibility Procedures become effective when:

a) The lowest RVR is less than 600 metres.
b) The cloud ceiling is 200 feet or less irrespective of the serviceability state of the ILS, lighting, standby power, e.t.c.

Pilots can expect ILS localizer and glide path signals to be fully protected from interference during the final approach, from the time that pilots are notified that LVP's are in operation until the time that pilots are notified that LVP's have been cancelled.

ATC will inform pilots when LVP's is in force. This may be obtained via the ATIS.

4.3 Localizer Sensitive Area (LSA)

For practicable purposes, the LSA is a rectangular area contained within parallel lines 150 m either side of the runway centreline and between the localizer aerial and the beginning of the runway demarcated by the airport authority.

No person, vehicle or aircraft is to be permitted to infringe the LSA from the time when:

a) An approaching aircraft is 10 NM from touchdown until it has vacated the LSA.
b) A departing aircraft has commenced its take-off run until it is airborne, and has passed the LOC antenna.

NOTE:
This means that landing or take-off clearance must not be issued if the LSA is known to be impaired.

4.4 Information to Pilots

In addition to the information normally transmitted by ATC, the following information must be passed by the appropriate controller to the pilot of every arriving aircraft: The current RVR readings for the landing runway.

4.5 Responsibility of Duty Airport Manager

The duty Airport Manager must notify all relevant instances including:

Avionicians

Electricians
Meteorological section
Police
All aircraft operators based on aerodrome
Fire Department
In-flight Catering Companies

Aerodrome Maintenance Personnel

i) That LVP's are in operation and to institute their respective ground safeguarding procedures.
ii) That LVP's are no longer in force.

4.6 Holding Points

In order to comply with the safeguarding requirements, aircraft awaiting take-off clearance must hold at the Category II holding positions, defined by painted taxiway markings, stopbar lights or at designated Category III holding positions, defined by stopbar lights and painted taxiway markings.

Above procedures are based on information contained in ECAC/CEAC document 17.

FAILED OR DOWNGRADED EQUIPMENT.		**EFFECT ON LANDING MINIMA.**	
		CAT II	**CAT III**
ILS(SEE NOTE 1).	ILS LOCALIZER DOWNGRADED TO CAT II.	NOT APPLICABLE.	NOT AUTHORISED.
	ILS LOCALIZER DOWNGRADED TO CAT I.	NOT AUTHORISED.	NOT AUTHORISED.
	ILS GLIDE PATH DOWNGRADED TO CAT II.	NOT AUTHORISED.	NOT AUTHORISED.
	ILS GLIDE PATH DOWNGRADED TO CAT I.	NOT AUTHORISED.	NOT AUTHORISED.
	ILS LOC AND/OR GP STANDBY TRANSMITTER U/S.	NOT ESSENTIAL.	NOT AUTHORISED.
	ILS TRANSMITTER STANDBY POWER SOURCE.	NOT AUTHORISED.	NOT AUTHORISED.
	OUTER MARKER.	USE ALTERNATIVE EG. LOCATOR NDB/ DME.	USE ALTERNATIVE MEANS EG LOCATOR NDB, DME.
	MIDDLE MARKER.	NOT REQUIRED.	NOT REQUIRED.
RVR (SEE NOTE 4)	TOUCHDOWN ZONE RVR NOT AVAILABLE.	NOT AUTHORISED.	NOT AUTHORISED.
	MIDPOINT RVR NOT AVAILABLE.	NOT AUTHORISED.	NOT AUTHORISED.

FAILED OR DOWNGRADED EQUIPMENT.		EFFECT ON LANDING MINIMA.	
APPROACH LIGHTS (SEE NOTES 1 & 3).	NO APPROACH LIGHTS.	NOT AUTHORISED.	FAIL OPERATIONAL AUTO LAND ONLY
	210M OF APPROACH LIGHTS AVAILABLE AS MEASURED FROM THRESHOLD.	NOT AUTHORISED	NO RESTRICTION.
	450M OF APPROACH LIGHTS AVAILABLE AS MEASURED FROM THRESHOLD.	NO RESTRICTION	NO RESTRICTION.
	SECONDARY POWER FOR APPROACH LIGHTS NOT AVAILABLE.	INCREASE RVR MINIMA TO 450M.	NO RESTRICTION.
RUNWAY LIGHTS (SEE NOTES 1 TO 4)	NO RUNWAY LIGHTS AVAILABLE (RWY EDGE THRESHOLD, END, CENTRE LINE AND TDZ LIGHTS U/S	NOT AUTHORISED	NOT AUTHORISED
	RUNWAY EDGE LIGHTS (INCLUDING THRESHOLD & END LIGHTS) NOT AVAILABLE. CENTRE LINE AND TDZ LIGHTS NORMAL	REQUIRED FOR MANUAL LANDINGS.	NOT REQUIRED.
	CENTRE LINE LIGHTS NOT AVAILABLE.	NOT AUTHORISED	NOT AUTHORISED
	TDZ LIGHTS NOT AVAILABLE.	DAY OPS AUTHORISED CLL MUST BE SERVICEABLE.	FULL OPS AUTO LAND ONLY. FAIL PASSIVE (DAY) AS FOR CAT II.
	SECONDARY POWER FOR RUNWAY LIGHTS NOT AVAILABLE. (ONLY REQUIRED FOR NIGHT OPERATION.	DAY OPS AUTHORISED IF RWY SURFACE MARKING CONTRAST IS GOOD.	NOT AUTHORIZED.
TAXIWAY LIGHTS.	TAXIWAY LIGHTING.	EDGE LIGHTS AND CL MARKS REQUIRED.	STOPBARS & CL LIGHTS REQUIRED. IF RVR IS LESS THAN 150M.
OTHER EQUIPMENT.	ANEMOMETER NORMALLY SERVING RWY IN USE.	USE OTHER SOURCE.	USE OTHER SOURCE.
	CEILOMETER.	NOT REQUIRED.	NOT REQUIRED.
	PAPI/VASI.	NOT REQUIRED.	NOT REQUIRED.

Note 1. In case of downgrading of ILS and deficiencies in lights determine first which approach category applies. Then check effect of lighting deficiencies for that category.

Note 2. For runway lights, multiple failures other than indicated in table are not approved.

Note 3. Deficiencies of approach and runway lights are treated separately and do not affect each other.

Note 4. Deficiencies in runway lights and RVR report, are not allowed together.

Note 5. 5% of any category of required lighting (ie. approach, TDZ, runway edge, centre line, taxiway lighting), may be inoperative, providing the pattern of lights is not distorted.

APPENDIX A

ABBREVIATION	DECODE
AIP SUP	Aeronautical Information Publication Supplement
ALT	Altitude
ATC	Air Traffic control
ATS	Air Traffic Service
C	Celsius
CAT	Category
CEAC	Common European procedures for the Authorization of CAT II and III operations
CLL	Centre Line Lights
DH	Decision Height
DA	Decision Altitude
EAT	Expected Approach Time
ECAC	European Civil Aviation Conference
ETA	Estimated Time of Arrival
FAF	Final Approach Fix
GCA	Ground Controlled Approach
HI	High Intensity
IAF	Initial Approach Fix
IAP	Instrument Approach Procedure
IAS	Indicated Airspeed
ICAO	International Civil Aviation Organisation
IFR	Instrument Flight Rules
ILS	Instrument Landing System
KT/KTS	Knot/Knots
LI	Low Intensity
LOC	Localizer
LSA	Localizer Sensitive Area
LVP	Low Visibility procedures
MAPt	Missed Approach Point
MDA	Minimum Descent Altitude
MDH	Minimum Descent Height
MET	Meteorology
MLS	Microwave Landing System
MSA	Minimum Sector Altitude

ABBREVIATION	DECODE
MSL	Mean Sea Level
NDB	Non Directional Beacon
NM	Nautical Mile
NOTAM	Notice to Airmen
OCA	Obstacle Clearance Altitude
OCH	Obstacle Clearance Height
OCT	Onward Clearance Time
OFZ	Obstacle Free Zone
PAR	Precision Approach Radar
RVR	Runway Visual Range
RWY	Runway
TAS	True Airspeed
TDZ	Touch Down Zone
THR	Threshold
VDP	Visual Descent Point
VFR	Visual Flight Rules
VMC	Visual Meteorological Conditions
VOR	Very High Frequency Omni-directional Range
VIS	Visibility

5 REDUCED VERTICAL SEPARATION MINIMUM (RVSM)

5.1 INTRODUCTION

Reduced Vertical Separation Minimum (RVSM) refers to vertical separation of aircraft by 1000 ft above Flight Level (FL) 290. Only State RVSM approved aircraft, will be permitted to operate in RVSM airspace within the South African Area of Responsibility (FAJA, FACA and FAJO FIRs).

After prior coordination with the appropriate ATC centre, State Aircraft (Military, Police and Customs) not RVSM approved will be cleared to operate within RVSM airspace within the South Africa Area of Responsibility, Two Thousand Foot (2000 ft) vertical separation will be applied to such aircraft.

Operators are to note that with the implementation of RVSM, FL 420 is no longer regarded as a valid Flight Level. The next available Flight Level will be FL 430.

In the event of a Two Thousand Feet (2000FT) separation being applied between aircraft within the RVSM flight level band, appropriate RVSM flight levels will be applied.

References: AFI Regional Supplementary Procedures (Doc 7030)ICAO Annex 2 Appendix 3 (a) refers.

5.2 OPERATION ABOVE FLIGHT LEVEL (FL) 410

Taking into consideration the extended distances of sectors which may be flown by operators,

allowance has been made for those non RVSM approved aircraft which are capable of operating at flight levels above FL410.

Restrictions placed on flights planning to operate at flight levels above FL410 require the climb to the planned flight level, once authorized and commenced under ATC clearance, to be conducted without interruption until clear of the RVSM flight level band. Conversely, once authorized and commenced under ATC clearance, descent from flight levels above FL410 are to be conducted without interruption until clear of the RVSM flight level band.

Non-RVSM approved aircraft intending to operate above FL 410 will need to flight plan in accordance with RVSM procedures of neighbouring Regions should the flight commence or terminate in those regions.

Reference: AFI Regional Supplementary Procedures (ICAO DOC. 7030)

5.3 DISCONTINUANCE OF OPERATIONS FOR NON RVSM APPROVED AIRCRAFT BETWEEN FLIGHT LEVEL (FL) 290 AND FLIGHT LEVEL (FL) 410 INCLUSIVE.

With the exception of State Aircraft non RVSM approved (Military, Police and Customs), all other non RVSM approved aircraft are permanently excluded from operations in the airspace between FL290 and FL410 inclusive.

Non RVSM approved aircraft are to plan for operations at FL280 or below, alternately above FL410.

5.4 IMPLEMENTATION OF REDUCED VERTICAL SEPARATION MINIMA (RVSM) WITHIN SOUTH AFRICAN AIRSPACE - Submission of Flight Plans for flight within RVSM airspace.

1) The following requirements when submitting flight plans for flight within the designated RVSM airspace are to be complied with.

2) The letter W shall be inserted in item 10 (Equipment) of the flight plan if the aircraft and operator have received RVSM State approval, regardless of the requested flight level. The aircraft registration shall be inserted in item 18 of the flight plan.
3) Operators of non-RVSM-approved State aircraft (Military, Police, Customs), with a requested flight level of 290 or above shall insert STS/NON RVSM in item 18 of the flight plan.

6 Aircraft lighting as a measure to prevent runway incursions

To reduce incidences of Runway incursions and to enhance situational awareness of an aircrafts intended movement, pilots should display aircraft exterior lights according to the guidelines stipulated below.

Prior to starting engines	Rotating Beacon on
Prior to taxi/push back	NAV lights on
Commencement of taxi	Taxi light on
Entering a runway for take off	Strobe light on
On receiving take off clearance	Landing light on
Crossing a runway	All above exterior lighting on (Including Strobe)

Pilots should endeavour to display those lights as are fitted to the aircraft as per the above guidelines however they may choose to make the aircraft more visible to other pilots or ground crew by any additional means or lighting, where the pilot in command deems it necessary (i.e. operating additional aircraft exterior lights on a congested Apron).

7 Callsign Appendices to annotate special handling requirements by ATC

Certain flights require specialised handling by Air Traffic Controllers to ensure optimum processing of the flight within the Air Traffic Management system. By using certain callsign appendices, the Air Traffic Controller is more aware of the type of flight and applicable service that should be provided to such flight. The use of the callsign appendix does not preclude the flight crew from completing normal notification on the Flight Plan and all required annotations that a flight requires special handling should still be made on the Flight Plan when this is filed before a flight. The use of the callsign appendix is an additional reminder of the requirement for special handling and in line with international best practice, flight crews should use the following callsign appendices when they wish to denote that the flight requires special handling.

Only the flight crew will use the callsign appendix and the ATC Unit handling the flight will not repeat the appendix of the callsign when issuing instructions to the flight.

The callsign appendix only needs to be used by the flight crew on first contact with the relevant ATC Sector.

a) Wake Turbulence

Heavy Wake Turbulence		
ACFT categorised as Heavy require ADDN spacing in front of other ACFT of lesser wake turbulence and incorrect spacing of a lighter ACFT behind a Heavy ACFT may result in the lighter ACFT encountering the wake turbulence of the heavy ACFT and be placed in danger.		
Callsign Appendix (Prefix or Suffix)	**Heavy**	**Suffix**
Flight Crew Actions	**ATC Actions**	
Flight crews of Heavy wake turbulence ACFT should use the suffix HEAVY on first contact with the following sectors; Clearance Delivery, Ground, Tower, Approach. It is not necessary to use the suffix with en- route/area sectors, as no additional spacing is required in these sectors.	ATC shall ensure that the correct departure, taxi route and runway allocation is issued to such ACFT. ATC shall also ensure that all following ACFT of lessor wake turbulence are adequately separated by the prescribed wake turbulence minima.	
Example	Tower, Airline 326 HEAVY.	

Super Heavy Wake Turbulence		
ACFT categorised as Super Heavy require ADDN spacing in front of other ACFT of lesser wake turbulence and incorrect spacing of a lighter ACFT behind a Super Heavy ACFT may result in the lighter ACFT encountering the wake turbulence of the heavy ACFT and be placed in danger.		
Callsign Appendix (Prefix or Suffix)	**Super**	**Suffix**
Flight Crew Actions	**ATC Actions**	
Flight crews of Super Heavy wake turbulence ACFT should use the suffix Super on first contact with the following sectors; Clearance Delivery, Ground, Tower, Approach. It is not necessary to use the suffix with en-route/area sectors, as no additional spacing is required in these sectors.	ATC shall ensure that the correct departure, taxi route and RWY allocation is issued to such ACFT. ATC shall also ensure that all FLW ACFT of lessor wake turbulence are adequately separated by the prescribed wake turbulence minima.	
Example	Tower, Airline992 Super.	

b) Medical Flights

Hospital Flights		
A Hospital flight is a medical flight that requires preferential routing, but can be subjected to a short delay provided that the delay is less than 10 minutes maximum. Usually an operator may use the callsign suffix Hospital when being dispatched to pick up a patient on a long route sector and the patient is not yet on board. The suffix may also be used when the patient is on board but not critical. However, flight crews are unable to determine the status of the patient unless a medical doctor on board has notified them of the patient's condition and is able to remain in direct contact with the flight crew. Hospital flights shall be annotated as STS/HOSP in field 18 of the Flight Plan. The purpose of the differentiation between Hospital and Medevac is that a controller shall still provide the Hospital flight with priority, but will not necessarily re-route other traffic that is marginally ahead or provides for a better sequence for a Hospital flight. A Hospital flight shall not normally be delayed for longer than 10 minutes maximum in holding. A Hospital flight may be upgraded to a Medevac flight at any stage in flight should the flight crew be advised that the patient's status is such that preferential treatment is needed.		
Callsign Appendix (Prefix or Suffix)	**Hospital**	**Suffix**
Flight Crew Actions	**ATC Actions**	
Flight crews of Medical flights which require priority handling but where a minimal delay can be accepted without having to re-route other aircraft shall use the prefix Hospital on first contact with all ATC Sectors. The term Hospital should not be used for ferry, test, training or positioning flights where there is no medical response needed.	ATC shall ensure that all Hospital flights receive the utmost priority, but shall not be required to re-route or position a Hospital flight ahead of another flight that may be marginally ahead or better positioned for the applicable sequence. A Hospital flight should not be delayed by more than 10 minutes as far as practical and if the delay is more than 10 minutes the flight crew shall immediately be advised thereof.	
Example	Approach, Airline574 Hospital.	

Medevac Flights		
A Medevac flight is a medical flight that requires preferential routing where no or minimal delay can be accepted due to the critical nature of the patient either on board or at the destination that the flight is routing to. Typically, the only time Medevac would be used by medical flights where no patient is on board is when responding to an accident scene or a case (in the case of EMS Helicopters or similar) or where a quick evacuation of the patient is required at the destination. Medevac flights require the highest level of preferential treatment and if necessary shall require other flights to be held or re-routed to ensure that no or minimal delay is afforded to the Medevac flight. Medevac flights shall be annotated as STS/MEDEVAC in field 18 of the Flight Plan.		
Callsign Appendix (Prefix or Suffix)	**Medevac**	**Suffix**
Flight Crew Actions	**ATC Actions**	
Flight crews of medical flights that require priority handling where no or minimal delay can be accepted due to the critical nature of the patient on board shall use the prefix Medevac on first contact with all ATC Sectors	ATC shall ensure that all Medevac flights receive the utmost priority and if necessary require other aircraft to be re-routed or held to accommodate the Medevac flight. A Medevac flight should not be delayed as far as practical and if a delay is inevitable the flight crew shall immediately be advised thereof.	

The Term Medevac should preferably only be used with a patient on board unless the flight is responding to an accident scene (i.e. EMS Helicopter) or the patient requires a quick evacuation from the destination aerodrome. In most other cases the term Hospital should be used when no patient is on board and the flight may be subjected to minor delays in sequencing. The Term Medevac should not be used for ferry, test, training or positioning flights where no medical response is needed.	
Example	Area, Airline786 Medevac.

c) Training Flights

Student Flights		
A Student flight is a flight where an ab-initio Student pilot is receiving flight instruction for a PPL (H-PPL). The prefix Student would be used by pilots holding a Student Pilot License only and not for pilots receiving conversion or higher training once having obtained a PPL.		
Callsign Appendix (Prefix or Suffix)	**Student**	**Prefix**
Flight Crew Actions	**ATC Actions**	
Student Pilots shall use the prefix Student on first contact with an ATC Sector to denote that ab-initio Student training is occurring on the flight for the acquisition of a PPL.	ATC shall be aware that the flight has an ab-initio student on board and where applicable shall provide progressive instructions or clearances.	
Example	Tower, Student ABC.	

Solo Student Flights		
A Solo Student flight is a flight where an ab-initio Student pilot has been sent solo in preparation for obtaining a PPL (H-PPL) and is doing solo circuits or GF work whilst still holding a Student Pilots License.The prefix Solo Student would only be used by pilots holding a Student Pilot License and not for pilots receiving conversion or higher training once having obtained a PPL		
Callsign Appendix (Prefix or Suffix)	**Solo - Student**	**Prefix**
Flight Crew Actions	**ATC Actions**	
Solo Student Pilots shall use the prefix Solo Student on first con-tact with an ATC Sector to denote that a Solo Student flight is occurring for the acquisition of a PPL.	ATC shall be aware that the flight has a student on board and where applicable shall provide progressive instructions or clearances.	
Example	Tower, Solo Student ABC.	

d) Flights that have Special Manoeuvring Characteristics

Helicopters		
A Helicopter is by nature more manoeuvrable than fixed wing aircraft; as such a helicopter can be requested to route at speeds and or routes not normally possible by fixed wing aircraft. Helicopters also have different VFR and minima from fixed wing aircraft and thus may be accommodated at lower minima than other fixed wing traffic.		
Callsign Appendix (Prefix or Suffix)	**Helicopter**	**Prefix**
Flight Crew Actions	**ATC Actions**	
A Helicopter pilot should consider using the prefix Helicopter on first contact with an ATC Sector to denote that the flight is being undertaken in a helicopter. Special consideration should be given in using this prefix when the registration marking does not make it easily discernable that the aircraft is a helicopter (i.e. not starting with H or R)	ATC shall provide routing and clearances as applicable to helicopter traffic.	
Example	Tower, Helicopter RBH.	

Gliders		
A glider is by nature less manoeuvrable than a powered fixed wing aircraft due to the fact that a glider cannot usually maintain an altitude fora sustained period of time. Gliders also have larger than normal wingspans and require special ground handling.		
Callsign Appendix (Prefix or Suffix)	**Glider**	**Prefix**
Flight Crew Actions	**ATC Actions**	
A glider pilot should consider using the prefix Glider on first contact with an ATC Sector or when broadcasting on a TIBA Frequency to de note that the flight is being undertaken in a glider. Special consideration should be given in using this prefix when the registration marking does not make it easily discernable that the ACFT is a helicopter (i.e. not starting with G)	ATC shall provide routing and clearances as applicable to glider traffic bearing in mind that an altitude may not be able to be maintained by the glider and that the glider may have to commence a number of turns whilst Thermalling to gain altitude.	
Example	Information, Glider GEL.	

Motor Gliders		
A Motor Glider may be as manoeuvrable as a powered fixed wing aircraft however the motor may not always be deployed on the glider and then aircraft will perform similar to a normal glider. Motor gliders have retractable power plant that when switched off shall normally fold away either into the nose cone or the fuselage of the aircraft. With the engine retracted the aircraft shall perform like a conventional glider and may not be able to maintain an altitude for a sustained period of time. Should the pilot need to deploy the engine again the time this takes for deployment and restart may vary depending on glider type.		
Callsign Appendix (Prefix or Suffix)	**Motor Glider**	**Prefix**
Flight Crew Actions	**ATC Actions**	

Motor Gliders	
A motor glider pilot should consider using the prefix Motor Glider on first contact with an ATC Sector or when broadcasting on a TIBA Frequency to denote that the flight is being undertaken in a Motor Glider.	ATC shall provide routing and clearances as applicable to glider traffic bearing in mind that an altitude may not be able to be maintained by the glider and that the glider may have to commence a number of turns whilst Thermalling to gain altitude.
Example	Information, Motor Glider GHC.

Hot Air Balloons		
A Hot Air Balloon is by nature less manoeuvrable than a powered fixed wing aircraft due to the fact that a balloon cannot usually control the direction of flight that it is moving in. The balloon pilot may be able to climb or descend however these commands shall take considerable time and depending on the altitude they are required to reach. Balloons also require special ground handling.		
Callsign Appendix (Prefix or Suffix)	**Balloon**	**Prefix**
Flight Crew Actions	**ATC Actions**	
A Hot Air Balloon Pilot should consider using the prefix Balloon on first contact with an ATC Sector or when broadcasting on a TIBA Frequency to denote that the flight is being undertaken in a Balloon.	ATC shall provide routing and clearances as applicable to balloon traffic bearing in mind that the direction of flight cannot be directly controlled by the pilot and is subject to wind and air movement.	
Example	Tower, Balloon BAL.	

Airships		
An Airship is by nature less manoeuvrable than a powered fixed wing aircraft due to the fact that the airship operates at slower speeds thana normal aircraft. The turning radius of an airship also differs vastly from that of a fixed wing aircraft. Although Airships are more manoeuvrable than Hot Air Balloons as they have full flight controls, they operate at speeds typically slower than most fixed wing traffic. Airships also usually require special ground handling.		
Callsign Appendix (Prefix or Suffix)	**Airship**	**Prefix**
Flight Crew Actions	**ATC Actions**	
An Airship pilot should consider using the prefix Airship on first contact with an ATC Sector or when broadcasting on a TIBA Frequency to de-note that the flight is being undertaken in an Airship.	ATC shall provide routing and clearances as applicable to Airship traffic bearing in mind that the speed and manoeuvrability of an air-ship differs from that of fixed wing traffic.	
Example	Tower, Airshop ZBC.	

e) Emergency Traffic

Mayday		
Usually when an aircraft declares an Emergency the radio transmission is started with the words Mayday, Mayday, Mayday however an air-craft that has been transferred from one frequency to another whilst still under the state of emergency should re-iterate that state that the aircraft has declared on first contact with each additional ATC Sector. Although ATCs are required to coordinate the emergency state prior to hand over the additional use also assists in allowing Emergency services and other traffic to identify the traffic requiring priority handling.		
Callsign Appendix (Prefix or Suffix)	**Mayday**	**Suffix**

Mayday	
Flight Crew Actions	**ATC Actions**
A Flight Crew that has declared and emergency shall on first contact with each new ATC Sector after declaring such, use the suffix Mayday with the callsign	ATC shall provide preferential routing and alerting services as required for the type of Emergency being experienced. Other traffic should be re-routed or held to accommodate the emergency traffic as far as practical.
Example	Approach, SEM Mayday...

Pan		
Usually when an aircraft declares a State of Urgency the radio transmission is started with the words Pan, Pan, Pan, however an aircraft that has been transferred from one frequency to another whilst still under the state of urgency should re-iterate that state that the aircraft has declared on first contact with each additional ATC Sector. Although ATCs are required to coordinate the Urgency state prior to hand over the additional use also assists in allowing Emergency services and other traffic to identify the traffic requiring priority handling.		
Callsign Appendix (Prefix or Suffix)	**Pan**	**Suffix**
Flight Crew Actions	**ATC Actions**	
A Flight Crew that has declared a state of Urgency shall on first contact with each new ATC Sector after declaring such, use the suffix Pan with the callsign.	ATC shall provide preferential routing and alerting services as required for the type of Emergency being experienced. Other traffic should be re-routed or held to accommodate the emergency traffic as far as practical.	
Example	Approach, WMN Pan...	

8 STRATEGIC LATERAL OFFSET PROCEDURES (SLOP) WITHIN THE SOUTH AFRICAN RNAV ROUTE STRUCTURE

8.1 Background

When Reduced Vertical Separation Minimum (RVSM) was implemented and the uptake of Global Navigation Satellite System (GNSS) became apparent, the risk of mid-air collision was increasing exponentially.

The obvious solution to reduce the vertical overlap risk in the system to meet the Target Level of Safety was to reduce the vertical overlap probability by recreating the randomness that RVSM and GNSS had removed from aircraft distribution. The result was the implementation of a lateral offset in increments of one mile.

Applying an offset to the right of track significantly reduces the collision risk between aircraft on same tracks and/or opposite tracks. The aircraft obtains lateral spacing from nearby aircraft that are immediately above and/or below.

8.2 Conditions

A flight crew should only apply SLOP if the following conditions are satisfied:

8.2.1 The appropriate ATS authority has authorised the application of SLOP in the airspace concerned and promulgated in aeronautical information publications (AIPs);

8.2.2 The aircraft is equipped with automatic offset tracking capability; and

8.2.3 The maximum allowed lateral offset for Phase I and II is detailed below. Only aircraft capable of offsetting in a fraction of a mile should apply SLOP.

NB - When operating in airspace where SLOP has been authorized, the decision to apply SLOP is the responsibility of the flight crew.

There is no ATC clearance required for this procedure and it is not necessary to advise ATC.

8.3 Application

Pilots are advised to use SLOP as standard operating procedure during normal operations on fixed RNAV route portions and FAJO airspace as described below to mitigate collision risk and wake turbulence;

8.3.1 Phase I

SLOP (Up to 0.5NM right of track) is approved on the following Bi-directional RNAV route portions listed below as published in the RSA AIP (ENR 3.3-12):

a) UQ21 (Between (ALENI – EGSOX)
b) UQ5 (Between (GEVIS – ESRET – RBV)
c) UQ17 (Between (ANVED – TGV)
d) UQ33 (Between (EPMAK – EKBOX – PEV)
e) UL211F (Between (APKIN – ITLIK)
f) UL224 (Between (APKIN – ITMEK)
g) UQ15 (Between (APKIN – OKDOG)

8.3.2 Phase II

SLOP (Up to 2NM right of track) is approved in FAJO airspace as published in the RSA AIP (ENR 2.1-37):

8.4 Conclusion

Aircraft utilising SLOP on the above mentioned RNAV route portions are to ensure that they immediately return to the nominal track of the specific RNAV's after exiting the areas of SLOP approval as listed in Phase I above.

Aircraft utilising SLOP in FAJO airspace are to ensure that they immediately return to the nominal track of RNAV/Routing's upon exiting FAJO as per adjoining airspace AIP requirements. Transiting from FAJO to FACA/FAJA airspace, the transition from 2NM SLOP to 0.5NM SLOP is required on the RNAV portions listed in Phase I above.